Fundamentals Chapter 7

Siddharth K

November 2024

1a.

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

1b.

$$\begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 8 & 7 \\ 5 & 4 \end{bmatrix}$$

1c.

$$\begin{bmatrix} 1 & 2 & 0 \\ 3 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 2 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 7 & 4 \\ 7 & 14 & 9 \\ 4 & 9 & 6 \end{bmatrix}$$

1d. First, we see that:

$$\begin{bmatrix} 1 & 2 & 0 \\ 3 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix} \left(\begin{bmatrix} 1 & 3 & 2 \\ 2 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \right)$$

$$= \begin{bmatrix} 1 & 2 & 0 \\ 3 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} x + 3y + 2z \\ 2x + 2y + z \\ y + z \end{bmatrix} = \begin{bmatrix} 5x + 7y + 4z \\ 7x + 14y + 9z \\ 4x + 9y + 6z \end{bmatrix}.$$

Then:

$$\left(\begin{bmatrix} 1 & 2 & 0 \\ 3 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 2 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix} \right) \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 & 7 & 4 \\ 7 & 14 & 9 \\ 4 & 9 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$= \begin{bmatrix} 5x + 7y + 4z \\ 7x + 14y + 9z \\ 4x + 9y + 6z \end{bmatrix},$$

which matches the first one.

2. The transformation represented by R_θ is a rotation counterclockwise by θ radians. This is because carrying out the multiplication

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

gives us

$$\begin{bmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{bmatrix}$$

which is the formula for rotation of the point (x, y) about the origin by θ radians.

3a. We can uniquely write $\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$. So thinking of $\mathbf{v}$ as $(a_1, \ldots, a_n)$ means uniquely representing $\mathbf{v}$ by its coordinates in $\mathbb{F}^n$ with respect to the given basis.

3b. Assume that $(a_1, \ldots, a_n)_V = (b_1, \ldots, b_n)_W$ for all $a_i, b_i \in \mathbb{F}$. Then any vector $\mathbf{v} \in \mathbb{F}^n$ has the same coordinates whether it's represented in the basis B_V or B_U. Now, let $(a_1, \ldots, a_n) = (b_1, \ldots b_n) = (0, \ldots, 0, 1, 0 \ldots, 0)$ where $a_i = b_i = 1$. By the assumption, we have that $\mathbf{v}_i = \mathbf{w}_i$. Doing this for all i gives us $\mathbf{v}_i = \mathbf{w}_i$ for all i.

Now, assume that $\mathbf{v}_i = \mathbf{w}_i$. There is only one way to write a vector $\mathbf{v}$ in terms of $\mathbf{v}_i$, so $(a_1, \ldots, a_n)_V = (b_1, \ldots, b_n)_W$.

4. Since $T(\mathbf{u}_i) = \mathbf{w}_i$, given any $\mathbf{w} \in W$, we have that

$$\mathbf{w} = \sum a_i\mathbf{w}_i = \sum a_i T(\mathbf{u}_i) = T\left(\sum a_i\mathbf{u}_i\right) = T(\mathbf{u})$$

for some $\mathbf{u} \in U$. So T is invertible.

5a. By the definition of matrix multiplication, $C = AB$ is defined as

$$c_{ij} = \sum_{l=1}^{n} a_{il}b_{lj}.$$

Since i and j range from 1 to n, there are n values for i and j in c_{ij}, so C is an $n \times n$ matrix.

5b. We have that the entry at (i, j) in AI_n is the total sum of each entry in $\begin{bmatrix} a_{i1} & \ldots & a_{in} \end{bmatrix}$ multiplied by each entry in $\begin{bmatrix} 0 & \ldots & 1 & \ldots & 0 \end{bmatrix}$ (where the 1 is at position j), which is just a_{ij}. So $A = AI_n$.

Also, we have that the entry at (i, j) in I_nA is the total sum of each entry in $\begin{bmatrix} 0 & \ldots & 1 & \ldots & 0 \end{bmatrix}$ multiplied by each entry in $\begin{bmatrix} a_{1j} & \ldots & a_{nj} \end{bmatrix}$ (where the 1 is at position i), which is just a_{ij}. So $A = I_nA$.

5c. First, consider the product $A(BC)$. We have that

$$A(BC)_{ij} = \sum_{k=1}^{p} a_{ik}(BC)_{kj} = \sum_{k=1}^{p} a_{ik}\left(\sum_{l=1}^{n} b_{kl}c_{lj}\right)$$

$$= \sum_{l=1}^{n} \left(\sum_{k=1}^{p} a_{ik} b_{kl}) c_{lj} \right).$$

Now, consider the product $(AB)C$. We have that

$$(AB)C_{ij} = \sum_{l=1}^{n} \left(\sum_{k=1}^{p} a_{ik} b_{kl}) c_{lj} \right),$$

which matches the expression for $A(BC)_{ij}$. Since this is true for all i, j, we have that $A(BC) = (AB)C$.

5d. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$. The condition is that (a, b) is not a scalar multiple of (c, d). This is because we need the system of equations created by $AB = I_2$ to have a unique solution, which is only true when (a, b) is not a scalar multiple of (c, d).

5e. Any matrix A that isn't a scalar multiple of I doesn't guarantee that $AB = BA$ for all 2×2 matrices B. To see this, we can consider a few cases for B. Let B equal $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$. Then

$$AB = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a & 0 \\ c & 0 \end{bmatrix},$$

and

$$BA = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix},$$

so we have that $b = c = 0$. This tells us that A is of the form $\begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix}$. Now, letting B equal $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ gives us

$$AB = \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & a \\ 0 & 0 \end{bmatrix},$$

and

$$BA = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} = \begin{bmatrix} 0 & d \\ 0 & 0 \end{bmatrix},$$

so $a = d$. This tells us that A is of the form $\begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}$. Furthermore, multiplying this by any 2×2 matrix B reveals that $AB = BA$.

6. The system of equations can be written as

$$A\mathbf{v} = \begin{bmatrix} 3 & -5 & 2 \\ -1 & 2 & 2 \\ 2 & -1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}.$$

If you know the inverse of A, you could multiply both sides by A^{-1}, and the result would be

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 & -5 & 2 \\ -1 & 2 & 2 \\ 2 & -1 & -1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}.$$

7. Lets say $U = \{\mathbf{u}_i\}, W = \{\mathbf{w}_i\}$ are bases for V. The matrix P_U^W maps each $\mathbf{u}_i$ to $\mathbf{w}_i$, so its inverse maps each $\mathbf{w}_i$ to $\mathbf{u}_i$. But this is exactly P_W^U, so $(P_U^W)^{-1}$ and P_W^U map each basis vector in W to the same one in U, which means they are the same transformation.

8. Reflexivity: $A = IAI$ so $A \sim A$.

Symmetry: $A \sim B$ implies that $A = PBP^{-1}$ for some matrix P. Multiplying both sides on the left by P^{-1} and on the right by P gives us $B = P^{-1}AP$, so $B \sim A$.

Transitivity: $A \sim B, B \sim C$ implies that $A = PBP^{-1}$ and $B = QCQ^{-1}$ for some matrices P, Q. Substituting the expression for B into A, we get $A = PQCQ^{-1}P^{-1}$. Since $(PQ)^{-1} = Q^{-1}P^{-1}$, we have that $A \sim C$.

9. Let $I_n = PAP^{-1}$. Multiplying both sides by P^{-1} on the left and P on the right gives us $A = P^{-1}P = I_n$. So $A = I_n$ is the only solution to $I_n = PAP^{-1}$.

10. We have that

$$\begin{bmatrix} 0 & 1/2 \\ 2 & 0 \end{bmatrix}^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I,$$

so $A^{10} = I$.

11. Consider $A\mathbf{v} = \mathbf{w}$ where the k'th column of A is all zeroes. Expanding the product shows that the result $\mathbf{w}$ is calculated in terms of every entry of $\mathbf{v}$ except v_k. Which means, if we wanted to map $\mathbf{w}$ to some unique $\mathbf{v}$, we couldn't because v_k can vary freely but $A\mathbf{v}$ will still equal $\mathbf{w}$. So A is not invertible.

Now, consider $A\mathbf{v} = \mathbf{w}$ where the k'th row of A is all zeroes. Since the k'th entry of $\mathbf{w}$ is the dot product of the k'th row and $\mathbf{v}$, it will always be 0. So any vector in W with a k'th entry other than 0 isn't reached by the linear transformation, and A is not invertible.

12. If $\ker(A)$ is nontrivial then A is not invertible. This is because 0 can then be reverse-mapped to more than one vector, which is a contradiction.

Let's say $\mathbf{v}_i$ is the basis that is being mapped to $\mathbf{w}_i$ (the columns of A) by the linear transformation, which we can call T. Since $\{\mathbf{w}_i\}$ is not linearly

independent, there exist nonzero a_i such that

$$\sum a_i \mathbf{w}_i = 0.$$

Since $\mathbf{w}_i = T(\mathbf{v}_i)$,

$$\sum a_i T(\mathbf{v}_i) = T\left(\sum a_i \mathbf{v}_i\right) = 0.$$

Since $a_i \neq 0$ and $\mathbf{v}_i$ are linearly independent, $\sum a_i \mathbf{v}_i \neq 0$. So $T(0) = 0$ and $T\left(\sum a_i \mathbf{v}_i\right) = 0$, which means the kernel is nontrivial and A is not invertible.